

Ilham Wahyu Saputro

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Semarang, Indonesia

Full Stack Developer with hands-on experience in designing and developing a web-based system for the detection and diagnosis of diabetic retinopathy, focusing on user-friendly interfaces and efficient data visualization. Skilled in integrating modern web technologies with machine learning models to deliver interactive and responsive applications. Currently developing an advanced sign language translation system utilizing Convolutional Neural Networks (CNN) combined with MediaPipe Hands

Experiences

organization experience– Universitas Stikubank	1 Okt 2022 – 30 Sep 2024
Social media admin – BriComp	1 Nov 2024 – 27 Feb 2025
Waiter – Hotways chicken pamularsih	10 Jan 2025 – 29 Jan 2026
Cashier – Hotways chicken pamularsih	30 Jan – 29 Apr 2026
Coding Camp powered by DBS Foundation – Dicoding	16 Feb – 11 jul 2025
- Completed 9 specialized Full Stack Developer modules from foundational concepts to advanced implementation under Accenture Full Stack Developer: Developing a web-based retinal detection system integrated with machine learning. Tech Stack: Express.js, Node.js, Three.js, React.js, MongoDB, Tailwind.css	
Company Profile Website Development (PT. Enggal Jaya) – Intern	
System Architecture: Developed a responsive company profile website using HTML, CSS, Bootstrap, and JavaScript, focusing on clean UI design and structured navigation across multiple pages (Home, Service, About, Contact). Integration & Features: Implemented real-time visitor tracking using Firebase Realtime Database and integrated Google Maps API for location visualization. Designed interactive UI components such as image sliders and service showcases to enhance user engagement. Tech Stack: HTML, CSS, Bootstrap, JavaScript, Firebase, Google Maps API.	
Full Stack Engineer – PT. Sydeco	13 Apr 2026 – Now
- Support developer for RitAPI, a Layer 6 AI-driven security engine: built Python detection modules (TLS, ASN, JSON Schema) and integrated Scikit-learn anomaly detection into the ARCHANGEL 2.0 MiniFW-AI ecosystem. - Productionized Python scripts into Linux services via systemd, managed .deb packaging pipelines, and orchestrated cross-platform environments (WSL/Windows) to ensure zero-downtime server deployment. - Developed an internal Flask dashboard for real-time security monitoring (alerts, threat logs, blocked IPs); simultaneously revamped the MiniFW-AI UI with sticky navigation, optimized spacing, and fully responsive cross-browser layouts. - Built and normalized a dataset for the Contract Risk Analyzer, enabling automated risk scoring to identify hazardous clauses in contracts.	

Education Level

Universitas Stikubank – Semarang	2022 - 2026
Bachelor of Informatics Engineering. GPA 3.77	

Technical Skills

Languages: Python, JavaScript, SQL
Frameworks & Libraries: bootstrap OpenCV, Flask, TensorFlow, NumPy, React (basic)
MLOps: Docker
DevOps & Infrastructure: GitHub Action, Linux WSL, Git

Certifications

Memulai Dasar Pemrograman untuk Menjadi Pengembang Software
Belajar Dasar Pemrograman Web
Belajar Dasar Pemrograman JavaScript
Belajar Back-End Pemula dengan JavaScript
Belajar Pengembangan Web Intermediate
Dicoding

Projects

Company Profile Website Development (PT. Enggal Jaya)

- **System Architecture:** Developed a responsive company profile website using HTML, CSS, Bootstrap, and JavaScript, focusing on clean UI design and structured navigation across multiple pages (Home, Service, About, Contact).
- **Integration & Features:** Implemented real-time visitor tracking using Firebase Realtime Database and integrated Google Maps API for location visualization. Designed interactive UI components such as image sliders and service showcases to enhance user engagement.
- **Tech Stack:** HTML, CSS, Bootstrap, JavaScript, Firebase, Google Maps API.

RetinaScan – Sistem Deteksi Dini Retinopati Diabetik untuk Penderita Diabetes

Coding Camp powered by DBS Foundation [Capstone Project]

- **Full Stack Developer:** Developing a web-based retinal detection system integrated with machine learning.
- **Tech Stack:** Express.js, Node.js, Three.js, React.js, MongoDB, Tailwind.css

Convolutional Neural Network dengan augmentasi data terkontrol untuk pengenalan huruf Bahasa Isyarat Indonesia (SIBI)

Thesis

- **Focus:** Real-Time Indonesian Sign Language (SIBI) Letter Recognition using Computer Vision.
- **Methodology:** Developed a Convolutional Neural Network (CNN) model trained on grayscale image data (64x64) with preprocessing techniques including normalization and resizing. Implemented real-time detection using OpenCV by integrating the trained model for live webcam-based prediction.
- **Result:** Achieved accurate real-time classification of SIBI alphabet gestures with confidence-based filtering, enabling efficient and responsive hand gesture recognition.
- **Tech Stack:** Python, TensorFlow/Keras (CNN), OpenCV, NumPy, Scikit-Learn.

RitAPI & ARCHANGEL 2.0 MiniFW-AI – Maintenance and Technical Support

Internal Project

- **Focus:** Maintained operational stability of an AI-driven Layer 6 API security engine and the MiniFW-AI ecosystem, encompassing bug fixing, system-level debugging, and front-end UI refinements.
- **Methodology:** Performed in-depth debugging on Python-based detection modules (TLS Certificate Analysis, ASN Trust Scoring, JSON Schema Validation) and Scikit-learn integration. Overhauled UI components including sidebar restructuring, sticky navigation, and responsive spacing, while resolving cross-browser visual inconsistencies through iterative testing.
- **Result:** Successfully stabilized the security system against production-level errors, ensured seamless anomaly detection, and delivered a cleaner, more consistent, and fully responsive user interface across all devices.
- **Tech Stack:** Python, Scikit-learn, JavaScript, HTML5, CSS3, Bootstrap.

Real-Time Security Monitoring Dashboard

Internal Project

- **Focus:** Developed an internal web-based dashboard to visualize real-time security metrics, enabling the operations team to monitor threats effectively.
- **Methodology:** Built a Flask web application integrated directly with the RitAPI detection backend. Designed dynamic views to display live alerts, threat logs, and blocked IP lists, prioritizing intuitive data readability for non-technical operational staff.
- **Result:** Delivered a reliable monitoring tool that significantly accelerated incident response times through clear and actionable data visualization.
- **Tech Stack:** Python, Flask, Jinja2, JavaScript, Chart.js, HTML/CSS.

Contract Risk Analyzer – Legal Dataset Engineering

Internal Project

- **Focus:** Built and normalized a trilingual (EN, ID, FR) dataset to support automated risk scoring for identifying hazardous clauses in legal contracts.
- **Methodology:** Designed a database comprising unique clause categories and constructed classified CSV repositories (dangerous, abusive, illegal, etc.).
- **Result:** Produced a high-integrity, duplicate-free dataset seamlessly integrated into the Contract Risk Analyzer MVP, enabling jurisdiction-aware legal compliance checking.
- **Tech Stack:** Python, Pandas, CSV, Excel.

